Missed cleavages

How completely the digest cut, read off the sequences of the peptides each sample actually produced.

AT A GLANCE

Where	Peptide Level → Missed Cleavage Plot
Needs	A peptide report, loaded as the Peptide level in Data → Upload, and an annotation. The peptide and precursor levels are read from the long-format report on disk rather than from a wide matrix, and this page needs an abundance in it: the report must carry the column Precursor.Quantity, which is fixed in the code and is not the **Abundance Column** offered on the quality-control pages. It also needs a stripped-sequence column and a file-name column.
Annotation	With a protein dataset loaded, the protein annotation is used. Without one, the peptide annotation made in Data → Annotation is. Until one of them exists the page says so and draws nothing.
Each bar	One sample, stacked to 100 %. Each segment is the share of the distinct *stripped* sequences — sequences without their modifications — detected in that sample that carry that many missed cleavages. Not precursors, not spectra, and not sites.
The rule	Cleavage after K and R, with proline blocking taken from a table in the code. Which of the two rules applies is chosen on the page with **Cleavage Rule**; it is not read from the search parameters.
Table	None.

1 What the plot answers

A digest that ran to completion leaves almost every peptide cut at every available site. This page counts the sites that survived, per peptide, and shows how the surviving counts are distributed in each sample. It is a figure about sample preparation: the enzyme, how much of it, how long, at what temperature, and how well the protein was denatured before it arrived.

Because each bar is normalised within its own sample, the comparison the figure supports is between the *shapes* of the bars, not their heights. A sample with a smaller zero-missed-cleavage segment has a larger fraction of detected sequences containing internal cleavage sites under the selected rule. This is consistent with less complete digestion, but search settings, sequence composition and detectability also affect the observed proportion.

> **A WORKED BAR**
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> Of 100 detected sequences, suppose 80 have no missed cleavage, 15 have one and 5 have two. The segments are 80 %, 15 % and 5 %; the share with at least one missed cleavage is 20 %. These describe the identified sequences under the chosen cleavage rule, not all molecules in the original digest.

Reading it

- One segment: the share of the sequences detected in that sample that carry that many missed cleavages. The segments of one bar sum to 100 % by construction.
- The legend, titled Number: the missed-cleavage counts present anywhere in the dataset, in ascending order from the foot of the bar upwards. A count nobody has gets no segment and no entry.
- With **Toggle Text** on, one percentage is printed near the foot of each bar: the share of that sample's sequences with *at least one* missed cleavage — the number usually quoted when a digest is assessed, and the sum of every segment above the zero segment rather than the segment it is printed over.
- The title names the rule the counts were made under, because the same peptides give different counts under the two rules and the figure would otherwise be ambiguous.

> **EASY TO GET WRONG**
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> The percentages are shares of what was *identified*, not of what was digested. A long peptide with three missed cleavages that the search never identified is absent from the report and therefore absent from the bar, so this is not an unbiased estimate of digestion completeness. It also says nothing about how many peptides are behind each bar: a sample with a tenth of the identifications of its neighbours still reaches 100 %. Read this page beside the coverage plot, which gives the counts.

2 Controls

Controls on this page

Control	What it does	Default	Where
Cleavage Rule	Which enzyme rule the count assumes: Trypsin or Trypsin / Lys-C. It changes the counts, not the data, and must match the digest that was actually run.	Trypsin	page
Toggle IDs	Adds the sample's own identifier under its condition_n label, on a second line.	off	page
Toggle Text	Prints, near the foot of each bar, the percentage of that sample's sequences with at least one missed cleavage, to one decimal.	on	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm and DPI between 72 and 600, and a very large figure has its resolution reduced so that the image stays under 24 megapixels.	25 cm × 15 cm, 100	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Value labels	The size of the percentage printed by **Toggle Text**. It is never set below 4.	8	appearance
Legend	Shown or hidden, and its title, size, colour, frame and position. Its entries are the missed-cleavage counts.	shown	appearance
Title, X axis, Y axis	Text, size, weight, colour and position of each label, and the size, rotation and colour of the tick labels. The plot writes Missed cleavages per sample followed by the rule in brackets, and Percentage of peptides with missed cleavage (%), and rotates the sample labels by 90°.	the plot's own	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Frame and grid	Which of the four spines are drawn, and the grid behind the bars.	all four spines	appearance
Background	Paper colour, transparency and the preset themes.	default	appearance

NOTE

There is no bar-colour control here. The segments take the drawing library's default colour cycle in the order of the missed-cleavage counts, so the colour of a segment depends on which counts are present in the dataset and is not fixed across datasets.

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

3 How it is computed

THE CLEAVAGE RULE

A missed cleavage is a cleavable residue *inside* a peptide. The rule is not read from the search parameters and is not inferred from the data: it is a table in the code, selected by **Cleavage Rule**.
Both rules cleave after lysine (K) and after arginine (R). They differ only in whether a following proline blocks the cut:

- Trypsin: K–P and R–P are both blocked, so neither counts.
- Trypsin / Lys-C: the Lys-C component is not inhibited by proline, so K–P counts; R is cut by trypsin alone, so R–P stays blocked.

For a sequence of residues $r_1 r_2 \ldots r_L$, read in upper case, the count is

$$\mathrm{mc} \;=\; \sum_{i=1}^{L-1} \mathbf{1}[\, r_i \in \{\mathrm{K}, \mathrm{R}\}\,] \cdot \mathbf{1}[\, r_i \text{ is not blocked by } r_{i+1}\,] \,.$$

The sum stops at $L - 1$: the last residue of a peptide is the expected cut site, not a missed one, and is never counted whatever it is. So SAMPLER has no missed cleavage, SAMKPLER has none under Trypsin and one under Trypsin / Lys-C, and SAMKPLEKR has one and two respectively.

FROM THE REPORT TO THE BARS

The long report is pivoted into a wide matrix: one row per distinct stripped sequence, one column per value of the file-name column, and in each cell the *largest* Precur-

sor.Quantity of the rows that share that sequence and that run, so modified and multiply charged forms of one sequence collapse into one number.
The columns are matched to the annotation and renamed condition_n, exact run names first and the trailing number in the name only as a fallback. A sequence counts as *detected* in a sample when its cell holds a finite number other than zero; zero is read as not detected.
Each distinct sequence is then given its missed-cleavage count once, from its string alone, and for every sample j and count k

$$\mathrm{pct}_{j,k} \;=\; 100 \times \frac{\#\{\, s \,:\, \mathrm{mc}(s) = k \text{ and } s \text{ detected in } j \,\}}{\#\{\, s \,:\, s \text{ detected in } j \,\}}.$$

The denominator is the sample's own number of detected sequences, which is why every bar reaches 100 % and why the bars carry no information about depth.

There is no fixed share above which a digest is called failed, and this manual does not name one. The number that matters is the one your own laboratory reaches on a digest that worked: compare a new run against that, and against the other samples in the same experiment, rather than against a value from elsewhere.

Those two are the rules the tool supports, and no other protease is planned. The rule is a setting of this page rather than a property of the dataset, so it has to be set to match the digest that was actually run — which is why the figure carries it in its title.

Edge cases

- Rows with an empty sequence are dropped before anything is counted.
- A sequence missing, non-numeric or exactly zero in a sample does not enter that sample's numerator or its denominator.
- A sample in which nothing is detected, and an annotated sample that matches no column in the report, are both drawn as an empty bar of height zero rather than being left off the axis.
- A single replicate in a condition is drawn like any other sample; nothing on this page aggregates within a condition.
- If nothing at all is detected anywhere, no figure is drawn and the page reports that no peptide intensities are available.

4 Exporting

The download reproduces the plot at the width, height and resolution set on the page, which is why those three controls do not change the figure on screen. Choose PDF for a figure that will be scaled or edited, PNG for a slide.

> **IN THE SUMMARY REPORT**
>
> The plot also appears in the summary report, under the same **Cleavage Rule** as the page, and its title states which rule that was.

Behind this page

Function	What it produces
missed_cleavage_plot	The stacked percentage bars, one per sample, under the selected cleavage rule
count_missed_cleavages	The count for one sequence, or for a column of them. The per-sample table behind the bars — Sample, Count, Occurrences, Total, Percentage — is built by peptide_missed_cleavage_summary.
normalize_protease	Resolves the written rule to trypsin or trypsin_lysc and refuses anything else